Topic 5: The Open Economy

Introduction to Economics II

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Based on Mankiw & Taylor, Chapters 28–29

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What Is an Open Economy?

An **open economy** is one that interacts freely with other economies in the world.

Opening the economy has consequences for both sides of our model:

Aggregate supply is affected by:

- Technology transfers (modifies A), relocation of capacity (modifies K)
- Migration flows (modifies L), adaptation of knowledge from abroad (modifies H)

Aggregate demand is affected because:

- It now includes net exports: $Y = C + I + G + NX$
- Investment can be financed with domestic *or* foreign saving
- Saving can be invested in domestic *or* foreign assets

Recall: Topics 3–4: In the long run, demand-side changes do not affect $\bar{Y}$.

Roadmap of Topic 5

1. **Net Exports & Exchange Rates**
   - Trade balance, nominal and real exchange rates, the NX curve
2. **Capital Flows, Identities & the Small Open Economy**
   - From the loanable funds market to NCO, the financial mirror, $S - I = NCO = NX$
3. **Equilibrium: The Two-Panel Model**
   - Loanable funds + Foreign exchange $\rightarrow$ determines ε^*
4. **The Model in Action: Four Experiments**
   - What happens when S, I, r^*, or $NX(\varepsilon)$ changes?
   - Budget deficits, saving/investment incentives, foreign fiscal policy, trade policy

Net Exports

Recall: Topic 1: GDP includes net exports: $Y = C + I + G + NX$

$$NX = X - M$$

$$\text{Net Exports} = \text{Exports} - \text{Imports}$$

Definitions:

- **Exports (X):** Goods and services produced domestically and sold abroad.
- **Imports (M):** Goods and services produced abroad and purchased domestically.
- **Net exports** or **trade balance** (NX): the difference $X - M$.

- $NX > 0$: **Trade surplus** (the country is a net seller of goods)
- $NX < 0$: **Trade deficit** (the country is a net buyer of goods)
- $NX = 0$: Balanced trade

What Determines Net Exports?

Net exports depend on several factors:

1. **Consumer preferences** for domestic vs. foreign goods.
2. **Prices** of goods at home and abroad.
   - If domestic goods become more expensive $\rightarrow X\downarrow$, $M\uparrow \rightarrow NX\downarrow$
3. **Exchange rates** (the price of domestic currency in terms of foreign currency).
4. **Incomes** at home (Y) and abroad (Y^*).
   - Higher $Y \rightarrow$ more imports $\rightarrow NX\downarrow$
   - Higher $Y^* \rightarrow$ more exports $\rightarrow NX\uparrow$
5. **Transport costs** and **trade policies** (tariffs, quotas, regulations).

At long run: $Y = \bar{Y}$ and $Y^* = \bar{Y}^*$ are fixed, so the key variable is the **real exchange rate**.

The Nominal Exchange Rate

Nominal exchange rate (e): the rate at which one currency trades for another.

Convention: e = units of foreign currency per unit of domestic currency.

$e = 1.10$ dollars per euro $\rightarrow$ with 1 you buy \$1.10.

- **Appreciation** ($e\uparrow$): your currency buys more foreign currency.
  - e: $1.10 \rightarrow 1.25$. Euro appreciated.
- **Depreciation** ($e\downarrow$): your currency buys less foreign currency.
  - e: $1.10 \rightarrow 0.95$. Euro depreciated.

When the euro appreciates, European goods become more expensive for Americans and American goods become cheaper for Europeans.

Using e to Compare Prices of a Single Good

A car costs 20,000 in Spain and \$22,000 in the US. Which is cheaper?

Convert the Spanish price to dollars: Price in $\$ = e \times P_{\text{car}}^{\text{Spain}}$.

Scenario	e (\$/)	**Spanish car in \$**	**Cheaper?**
Weak euro	0.90	\$18,000	Spain
Neutral	1.10	\$22,000	Same
Strong euro	1.25	\$25,000	USA

Higher $e \rightarrow$ Spanish goods more expensive for Americans.

e works for comparing **one specific good**. But for aggregate trade we need to compare **entire baskets of goods** — and that requires accounting for the price levels of both countries.

From e to ε: Accounting for Prices

e compares *currencies*. But trade depends on the relative cost of *goods* — and that depends on prices too.

Example: e stays at 1.10. But Spanish inflation rises 20%. The car that cost 20,000 now costs $24{,}000 \times 1.10 = \$26{,}400$. Spanish goods got more expensive *without e moving at all*.

The problem is that e alone misses what happens to prices. The fix is simple — adjust e for the price levels of both countries:

Real exchange rate: $$\varepsilon = \frac{e \times P}{P^*}$$

ε is just e adjusted for the price levels of both countries.
It measures the relative price of the domestic basket of goods.

Example: $e = 1.10, \quad P = 120, \quad P^* = 100$: $\varepsilon = \frac{1.10 \times 120}{100} = 1.32$. European goods are 32% more expensive.

If instead $P = 80$: $\varepsilon = \frac{1.10 \times 80}{100} = 0.88$. European goods are 12% cheaper.

Real Appreciation and Depreciation

What matters is not the level of ε but the **direction of change**:

Real appreciation ($\varepsilon\uparrow$)	**Real depreciation** ($\varepsilon\downarrow$)
Domestic goods become *more* expensive.	Domestic goods become *cheaper*.
Exports fall, imports rise. $NX\downarrow$	Exports rise, imports fall. $NX\uparrow$

Since $\varepsilon = eP/P^*$, three things can cause a real appreciation: (1) nominal appreciation ($e\uparrow$), (2) domestic inflation ($P\uparrow$), (3) foreign prices fall ($P^*\downarrow$).

In rates of change (where $\hat{x} = \frac{\Delta x}{x}$ denotes the **growth rate** of x), the same definition gives:

$$\hat{\varepsilon} = \hat{e} + \pi - \pi^*$$

$$\text{growth rate of } \varepsilon = \text{growth rate of } e + \text{domestic inflation} - \text{foreign inflation}$$

This is not a new formula — it is the definition $\varepsilon = eP/P^*$ expressed in growth rates.

What Moves ε in Practice?

The formula $\hat{\varepsilon} = \hat{e} + \pi - \pi^*$ has important implications:

- Even if $\hat{e} = 0$: if $\pi > \pi^* \rightarrow \hat{\varepsilon} > 0 \rightarrow$ real appreciation. Competitiveness erodes *silently*, through prices alone.
- A nominal depreciation ($\hat{e} < 0$) can be offset if $\pi > \pi^*$. The country devalues its currency, but its prices rise just as fast.
- With flexible exchange rates, e tends to absorb inflation differentials in the long run. But in the short run, e moves for many other reasons (capital flows, monetary policy, speculation), so e and ε can diverge significantly.

$\rightarrow$ Topic 6 will study what determines π (the quantity theory of money).

When e Cannot Adjust: The Eurozone

In a **currency union**, there is **no nominal exchange rate** between member countries — they use the same currency. The variable e simply does not exist as a mechanism of adjustment.

This means that if one country has higher inflation than another, its products become more expensive and **nothing compensates**. All changes in competitiveness come through prices alone.

Spain vs. Germany (2000–2008):

- Spanish inflation was systematically higher than German inflation.
- Same currency, so no nominal adjustment is possible.
- Result: Spanish products became progressively more expensive relative to German ones — a *silent* real appreciation, year after year.
- This loss of competitiveness was one of the causes of the eurozone crisis.

In a currency union, ε is the *only* relevant measure of competitiveness.
There is no e between members to do the adjusting.

From Determinants to a Function

In slide 5 we listed five determinants of net exports. Now we formalise this:

$$NX = NX\Big(\underset{-}{\varepsilon},\ \underset{-}{Y},\ \underset{+}{Y^*},\ \text{preferences},\ \text{trade policy}\Big)$$

- $\varepsilon\uparrow \rightarrow$ our goods more expensive $\rightarrow X\downarrow,\ M\uparrow \rightarrow NX\downarrow$ (sign: $-$)
- $Y\uparrow \rightarrow$ we import more $\rightarrow NX\downarrow$ (sign: $-$)
- $Y^*\uparrow \rightarrow$ foreigners buy more from us $\rightarrow NX\uparrow$ (sign: $+$)

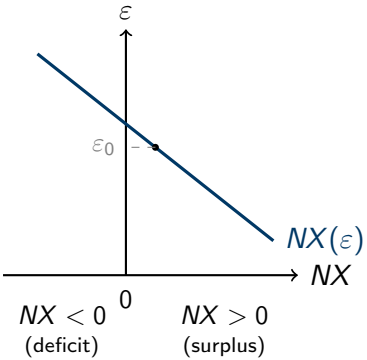
In the long run:

- $Y = \bar{Y}$ and $Y^* = \bar{Y}^*$ are fixed (determined by factors of production, Topic 3).
- Preferences and trade policy are taken as given.
- The only variable that adjusts is ε.

Therefore: $NX = NX(\varepsilon)$, with $\dfrac{dNX}{d\varepsilon} < 0$.

The Net Exports Curve

This is simply the **graphical representation** of what we already know: when ε rises, domestic goods become more expensive and net exports fall.



- Above ε_0: goods too expensive, imports exceed exports $\rightarrow$ deficit.
- Below ε_0: goods cheap, exports exceed imports $\rightarrow$ surplus.
- At ε_0: $NX = 0$ (trade is balanced).

Important: this curve gives us a *relationship* between ε and NX. To find the equilibrium, we need a second piece of information — the level of NX.

The NX Curve Is Not Enough

We have the curve $NX(\varepsilon)$: for every value of ε, it tells us the level of NX.

But which ε actually prevails? The curve alone cannot tell us — just like a demand curve alone cannot tell you the price without knowing supply.

We need a second piece of information: what is the level of NX in equilibrium?

To find this, we need to look at the **financial side** of the open economy. When a country trades goods, it also trades financial assets. These financial flows, which come from the loanable funds market we studied in Topic 4, will give us the missing piece.

Piece 1 (we have it): the curve $NX(\varepsilon)$ — how NX reacts to ε.

Piece 2 (we need it): a specific number for NX — the level.

Where the curve hits that number $\rightarrow$ that is ε^*.

The Small Open Economy and the Loanable Funds Market

Recall: Topic 4 (closed economy): $S = I$. The interest rate r adjusts to equilibrate the market. All saving stays at home.

Now we open the door. We model a **small open economy (SOE)**:

- **Small:** our economy cannot affect world markets.
- **Perfect capital mobility:** no barriers to financial flows.
- **Consequence:** $r = r^*$ (exogenous, set by world capital markets).

At r^*, saving and investment may not be equal: $S(r^*) \neq I(r^*)$.

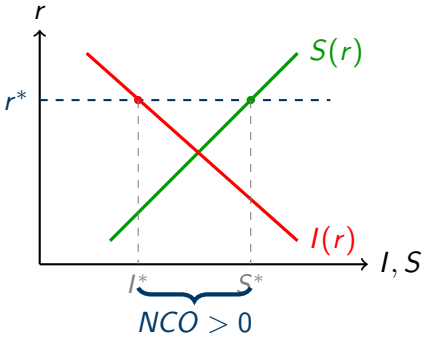
What do savers do with the excess? *Exactly what they did in Topic 4* — they lend in exchange for a return — but now they can lend **abroad**: buy foreign bonds, deposit in foreign banks, invest in foreign companies.

Net Capital Outflow (NCO) = purchase of foreign assets by domestic residents − purchase of domestic assets by foreigners.

$$S = I + NCO$$

Saving finances domestic investment *and* foreign lending.

When $S > I$: The Country Lends Abroad



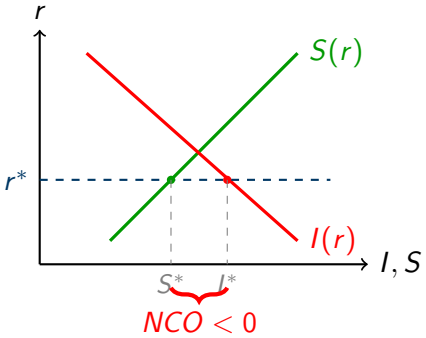
At this r^*, savers want to lend **more** than firms want to borrow.

In a closed economy, r would fall until $S = I$. But the door is open: r stays at r^*.

The excess saving flows **abroad**: savers buy foreign bonds, deposit in foreign banks, invest in foreign companies.

$S > I \Rightarrow NCO > 0$
The country is a **net lender**.

When $S < I$: The Country Borrows from Abroad



Now r^* is low. Firms want to invest **more** than domestic savers can finance.

The shortfall is covered by **foreign saving**: foreign investors buy domestic bonds, lend to domestic banks, invest in domestic companies.

$S < I \Rightarrow NCO < 0$
The country is a **net borrower**.

Key: The level of r^* determines whether the country lends or borrows.

From Goods to Finance: The Easy Direction

Example: Seat sells cars to the US for \$25 million.

1. Cars cross the border: Spain $\rightarrow$ US. This is an export ($NX\uparrow$).
2. Seat receives \$25M. Those dollars are a *claim on the US economy*.
3. Whether Seat keeps them in a US bank, buys US Treasury bonds, or converts them to euros (in which case another agent ends up holding the dollars) — a foreign financial asset exists somewhere ($NCO\uparrow$).

Goods $\rightarrow$ Finance: Every export automatically creates a foreign asset. Every import automatically creates a domestic asset held by foreigners.

This direction is *automatic*, transaction by transaction.

From Finance to Goods: The Subtle Direction

Example: A Spanish pension fund decides to buy US Treasury bonds and Toyota shares. No goods are involved — only financial assets.

So where are the exports? The connection is **indirect**, through ε:

1. The fund needs dollars and yen. It sells euros on the foreign exchange market.
2. If many Spanish savers do the same, there is selling pressure on the euro.
3. The euro depreciates: $\varepsilon \downarrow$.
4. Spanish products become cheaper for foreigners $\rightarrow$ exports rise, imports fall $\rightarrow$ $NX \uparrow$.

Finance $\rightarrow$ Goods: Financial outflows push ε down, which makes domestic goods cheaper, which raises NX.

NX rises until it equals NCO.

$NCO = NX$: Every Transaction Has Two Sides

Every international transaction is recorded **twice**: once in the trade account, once in the financial account.

Example: Seat sells cars to the US for \$25M. Trade account: export ($NX\uparrow$). Financial account: Seat holds \$25M in a US bank ($NCO\uparrow$). Same transaction, two books.

When you sum all transactions in a year: $NCO = NX$, necessarily. This is accounting, not a theory.

Now combine this with what we know from slides 15–17:

$$S = I + NCO \quad \text{and} \quad NCO = NX \quad \Longrightarrow \quad NX = S - I$$

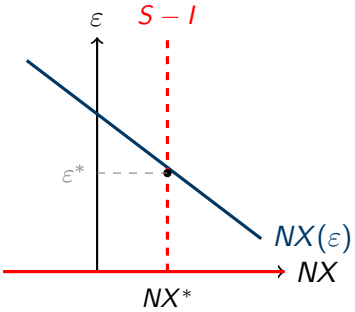
Since r^* determines S and I, $S - I$ is a **specific number**.

Example: If $S = 200$bn and $I = 150$bn, then $NX = 50$bn.

This is our missing piece. The financial side of the economy gives us a concrete number for NX. Not a curve, not a relationship — a number.

Two Pieces, One Equilibrium

Now we have everything we need:



Piece 1: The curve $NX(\varepsilon)$ from slide 13. For each ε, it gives a value of NX.

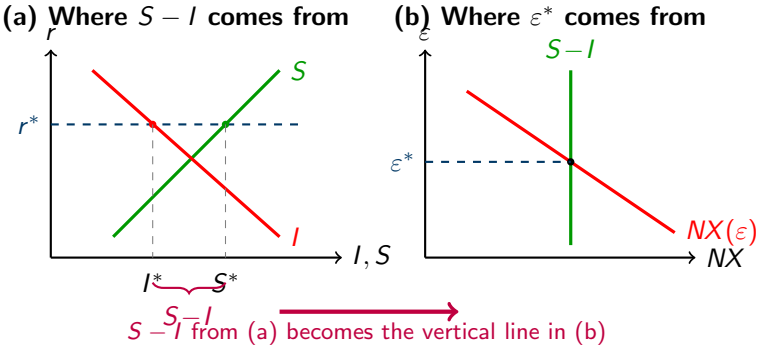
Piece 2: The number $S - I$ from slide 20. It tells us what NX *must* be.

Equilibrium: The ε at which the curve produces exactly $S - I$. That is ε^*.

This is the entire model.

The Two-Panel Model

The graph on slide 21 showed the idea. Now we add the **first step**: where does $S - I$ come from?



- **Panel (a):** At r^*, we find S and I $\rightarrow$ their difference is $S - I$ (a number).
- **Panel (b):** That number becomes a vertical line. Where it crosses $NX(\varepsilon)$ gives ε^*.

How the Model Works: Step by Step

Step 1: r^* is given by the world. At that r^*, find S and I from panel (a). Compute $S - I$.

Step 2: Take $S - I$ to panel (b). The curve $NX(\varepsilon)$ slopes down. The ε at which the curve hits $S - I$ is the equilibrium: ε^*.

Step 3: At ε^*, all identities are satisfied: $NX = S - I = NCO$. ✓

This is the formal version of the mechanism from slides 18–19: financial flows determine the level of NX, and ε adjusts until the goods market produces exactly that level.

What determines what:

r^* (from the world) $\rightarrow S - I$ (the level of NX) $\rightarrow \varepsilon^*$ (the price that makes it happen).

The Model in Action: Four Experiments

The model says $NX = S - I$ and ε adjusts. What happens when something changes?

There are only **four things** that can move:

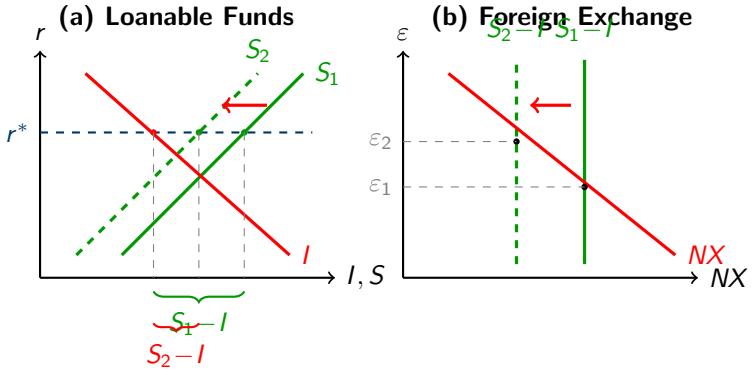
1. **S changes** — e.g. budget deficit ($S\downarrow$) or saving incentives ($S\uparrow$).
2. **I changes** — e.g. investment tax credits ($I\uparrow$).
3. **r^* changes** — e.g. the rest of the world runs a fiscal deficit ($r^*\uparrow$).
4. **The curve $NX(\varepsilon)$ shifts** — e.g. tariffs or import quotas.

Cases 1–3 change $S - I \rightarrow$ the vertical line in panel (b) moves $\rightarrow$ both NX and ε change.

Case 4 is different: $S - I$ does not change $\rightarrow$ only ε changes. This is the *surprising* result.

When S Falls: Budget Deficit

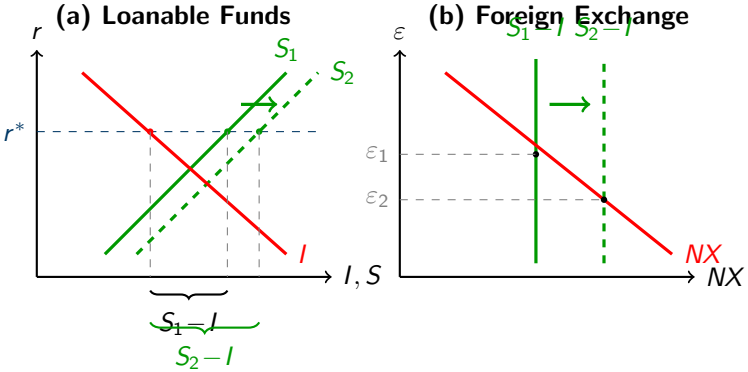
Policy: The government runs a budget deficit ($G\uparrow$ or $T\downarrow$) $\rightarrow S_{\text{public}}\downarrow \rightarrow S\downarrow$.



Results: $S\downarrow \rightarrow S-I\downarrow \rightarrow \varepsilon\uparrow$ (appreciation) $\rightarrow NX\downarrow$. Investment unchanged ($r=r^*$). This is why budget deficits and trade deficits often go together (*twin deficits*).

When S Rises: Saving Incentives

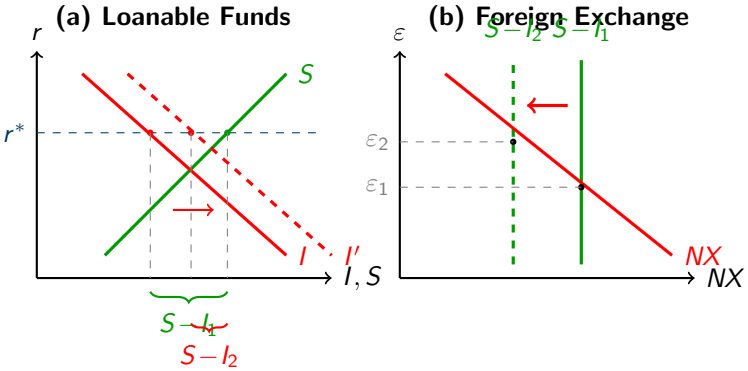
Policy: Tax incentives that increase private saving.



Results: $S\uparrow \rightarrow S - I\uparrow \rightarrow$ vertical line shifts right $\rightarrow \varepsilon\downarrow$ (depreciation) $\rightarrow NX\uparrow$.
The trade balance improves. Investment does not change (since $r = r^*$).

When I Rises: Investment Incentives

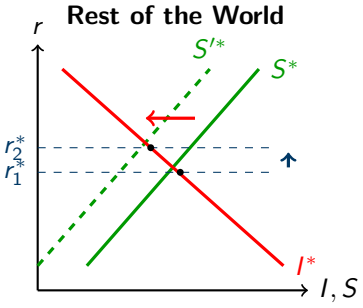
Policy: Tax incentives that increase domestic investment demand.



Results: $I\uparrow \rightarrow S - I\downarrow \rightarrow$ vertical line shifts left $\rightarrow \varepsilon\uparrow$ (appreciation) $\rightarrow NX\downarrow$.
Investment rises, but net exports fall. Future growth improves (more K).

When r^* Rises: Fiscal Policy Abroad (I)

What if the rest of the world runs a budget deficit?



The foreign government runs a deficit $\rightarrow S^*$ falls.

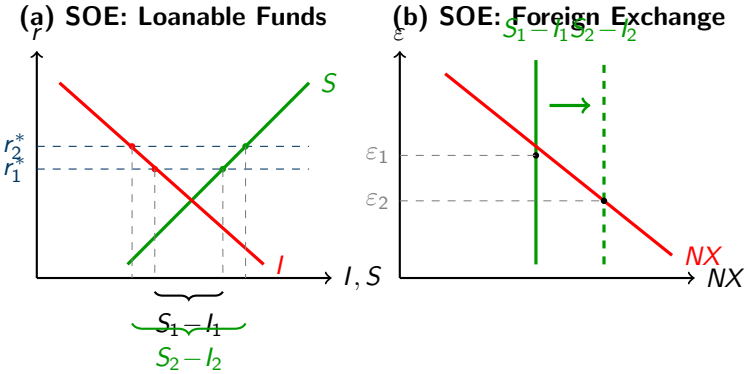
In the world loanable funds market, less saving means a higher equilibrium interest rate.

r^* rises from r_1^* to r_2^*.

What does this mean for our SOE?

When r^* Rises: Fiscal Policy Abroad (II)

At the new, higher r_2^*: domestic saving rises and domestic investment falls $\rightarrow S - I$ increases.



Results: $r^* \uparrow \rightarrow S \uparrow$, $I \downarrow \rightarrow S - I$ increases $\rightarrow \varepsilon \downarrow$ (depreciation) $\rightarrow NX \uparrow$.

When $NX(\varepsilon)$ Shifts: Trade Policy

Can trade restrictions improve the trade balance?

Common protectionist policies:

- **Tariff:** Tax on imported goods.
- **Import quota:** Limit on quantity of imports.
- **Other barriers:** Regulations, standards, subsidies to exporters.

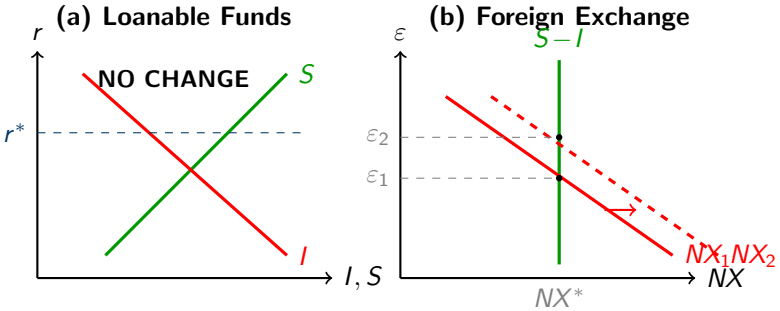
The political argument:

- "We import too much — let's restrict imports to fix the trade deficit!"

Direct effect: At any given ε, imports decrease $\rightarrow$ NX curve shifts rightward.

But what happens in general equilibrium?

Effect of Import Restrictions



Results:

- Trade restrictions do not affect S or $I \rightarrow S - I$ is unchanged.
- NX shifts right, but ε rises (appreciation) until NX returns to its original level.
- **Net exports are unchanged!** Only ε changes.

Why Trade Policy Doesn't Fix the Trade Balance

The trade balance is determined by $S - I = NX$.

Trade policy does not affect S or I!

What actually happens when imports are restricted:

1. Imports of protected goods decrease.
2. But the real appreciation makes exports less competitive $\rightarrow$ exports also decrease.
3. Total volume of trade $(X + M)$ falls, but NX is unchanged.

Consequences:

- Protected industries benefit, but exporters are harmed.
- Consumers face higher prices and less variety.
- Overall efficiency loss: society loses gains from trade.

To improve the trade balance: must change S or I, not trade policy.

Summary: The Model in Action

What changes?	$S-I$	ε	NX
Budget deficit ($S\downarrow$)	$\downarrow$	$\uparrow$	$\downarrow$
Saving incentives ($S\uparrow$)	$\uparrow$	$\downarrow$	$\uparrow$
Investment incentives ($I\uparrow$)	$\downarrow$	$\uparrow$	$\downarrow$
Foreign fiscal expansion ($r^*\uparrow$)	$\uparrow$	$\downarrow$	$\uparrow$
Trade policy (tariffs)	$=$	$\uparrow$	$=$

Pattern: If $S-I$ changes, both ε and NX change. If only the $NX(\varepsilon)$ curve shifts, NX does not change — only ε adjusts.

Key difference with closed economy: In a SOE, fiscal policy crowds out *net exports* rather than *investment* (because r is fixed at r^*).

The Classical Dichotomy in an Open Economy

In the long run, real variables are determined independently of the quantity of money:

Factor markets:

$$L^d = L^d\left(\frac{W}{P}, K, A\right)$$
$$L^s = L^s\left(\frac{W}{P}\right)$$
$$K^d = K^d\left(\frac{R}{P}, L, A\right), \quad K^s = \bar{K}$$
$$\Longrightarrow \frac{W}{P},\ \frac{R}{P},\ L,\ K,\ \bar{Y}$$

Goods & financial markets:

$$\text{DA:} \quad Y = C(r^*) + I(r^*) + G + NX(\varepsilon)$$
$$\text{OA}_{\text{LP}}: \quad Y = \bar{Y}$$

The interaction between OA and DA determines ε^*.

- $\bar{Y}$ determined by supply (factors).
- $r = r^*$ by international markets.
- ε^* by $S - I = NX(\varepsilon)$.

The real exchange rate depends only on real factors in the long run.

→ Topic 6 (next): nominal variables (P, W, R, nominal e) will be determined by the money supply.

Summary: Key Concepts of Topic 5

1. **Open economy:** interacts with the rest of the world through goods, services, and capital flows.
2. **Exchange rates:** Nominal (e, currency price) and Real ($\varepsilon = eP/P^*$, relative goods price). Net exports depend on ε.
3. **Fundamental identities:** $S = I + NCO$ and $NCO = NX$ (always).
4. **Small open economy:** $r = r^*$; the trade balance is determined by $S - I$.
5. **Two-market model:** Loanable funds determines r and $S - I$; foreign exchange determines ε.
6. **Trade policy:** Does *not* affect NX. Only changes ε.